Scatter Plots and Bivariate Association — Making and Qualifying Predictions

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

1. 85	4. 65, 3	7. 93	10. 36
2. 22	5. —	8. —	11. 70.78
3. 69	6. 72.7	9. 36	12. —

Curriculum

Level: Grade 8 / Pre-Algebra
8.SP.A.1 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

3. *If they got 75.8: read the slope as the prediction and added the study time to it instead of multiplying*
9. *If they got 40: subtracted the two study times instead of the two quiz scores*
11. *If they got 63.7: removed the unusual score from the total but still divided by all ten students*

The habit this sheet builds: a trend line will return a number for any input you hand it, including inputs the data never covered and outputs the quiz cannot award. Computing the prediction is the easy half; the graded half is the sentence that says how far the prediction can be trusted. Full credit on the explanation problems needs a reason tied to *this* plot — its range, its one unusual point, or the gap between association and cause — not a general remark about being careful.

1. Reading one point off the plot.

Find 45 on the horizontal axis, go straight up to the dot, then across to the vertical axis. The dot sits level with $\boxed{85}$.
Note this is a *read*, not a prediction: this student is in the data. The trend line at 45 minutes gives $0.8(45) + 45 = 81$, so the line runs 4 points below what this student actually scored — normal scatter, not an error.

2. The highest score, and whether it fits.

Scan the plot for the highest dot, then read *down* to the axis rather than guessing from the trend. The top dot sits at a study time of $\boxed{22}$ minutes.
Does it fit the trend? No. Every other dot climbs steadily from left to right, and the trend line predicts $0.8(22) + 45 = 62.6$ at that study time — more than 27 points below the score this student earned. It is an **outlier**: a real data point that does not follow the pattern of the rest. It is worth noticing early, because it moves the mean (problems 6 and 11) and it weakens any claim built on the trend alone.

3. Predicting inside the data range.

The prediction is just the trend line evaluated at the given study time. Substitute $x = 30$ into $y = 0.8x + 45$:

$$y = 0.8(30) + 45 = 24 + 45$$

The predicted score is $\boxed{69}$.
This prediction is **interpolation**: 30 minutes lies inside the 10-to-50-minute range the class actually studied, so the plot is real evidence for it.
Watch for: adding instead of multiplying, $0.8 + 30 + 45 = 75.8$. The slope is a rate per minute, so it multiplies the minutes.

4. A prediction and its residual.

(a) Substitute $x = 25$ into the trend line:

$$y = 0.8(25) + 45 = 20 + 45$$

The predicted score is $\boxed{65}$.
(b) A residual is always *actual minus predicted*, in that order, so the sign carries meaning:

$$68 - 65$$

The residual is $\boxed{3}$ points.
Because the residual is **positive**, the student scored above the line: the line *under*-predicted this student by 3 points. A negative residual would have meant the opposite. Residuals are how you measure whether a line fits, one point at a time.

5. Why association is not cause.

A scatter plot shows only that two quantities move together in this class's data. It cannot rule out a third factor that moves both, and it cannot show which direction any influence runs.
A full-credit answer names *one* reason and *one* plausible alternative factor. Reasons that earn credit: the data was observed, not from an experiment where study time was assigned; nobody controlled anything else; a pattern in ten students is not evidence about students in general. Alternative factors that earn credit: students who already understand the material both study more comfortably and score higher; sleep; tutoring; interest in the subject; test-taking practice.
Reversal is also worth accepting: students who expect to do well may be the ones willing to sit and study, so the arrow could point the other way.
Open explanation — $\boxed{\text{open response}}$; check that a specific alternative factor is named, not just the phrase "correlation is not causation".

6. The mean score of all ten students.

Add every score on the plot, then divide by the number of dots:

$$52 + 58 + 61 + 90 + 68 + 70 + 76 + 79 + 85 + 88 = 727$$

$$\text{mean} = \frac{727}{10}$$

The mean is $\boxed{72.7}$ points.

Which way does the outlier pull? Up. The 90 sits far above the trend at its own study time, and it is the only score that does. Problem 11 measures the pull exactly: the mean drops to about 70.78 without it, so that single student lifts the class mean by roughly 2 points.

7. Predicting outside the data range.

The arithmetic is identical to problem 3 — substitute $x = 60$:

$$y = 0.8(60) + 45 = 48 + 45$$

The trend line predicts $\boxed{93}$.

The number is right; the confidence is the issue, and that is problem 8. Nobody in this class studied 60 minutes, so the plot contains no evidence that the line still holds there.

8. Interpolation versus extrapolation.

Interpolation means predicting at an input *inside* the range the data covers. Study times on the plot run from 10 to 50 minutes, so the 30-minute prediction of 69 sits well inside that range, with real dots on both sides of it supporting the line.

Extrapolation means predicting at an input *outside* that range. The 60-minute prediction of 93 is extrapolation: it assumes the straight-line pattern keeps going past the last dot, and nothing in the data shows that it does. Real score-versus-study relationships usually flatten out — there is only so much a quiz can measure, and a score cannot pass 100 at all.

Full credit needs both words used correctly *and* the range 10 to 50 cited as the reason. Open explanation — $\boxed{\text{open response}}$.

9. Comparing two students' scores.

The question asks about *scores*, so read both dots on the vertical axis and subtract those, not the study times:

$$88 - 52$$

The 50-minute student scored $\boxed{36}$ points higher.

Watch for: subtracting the study times, $50 - 10 = 40$. That is a difference in minutes, not in points — a units error that looks plausible because the numbers are close.

10. Where two trend lines agree.

Two lines predict the same score at the study time where their outputs are equal, so set the expressions equal and solve:

$$0.8x + 45 = 1.3x + 27$$

$$45 - 27 = 1.3x - 0.8x \quad \Rightarrow \quad 18 = 0.5x$$

The two classes agree at $\boxed{36}$ minutes.
Check: first class $0.8(36) + 45 = 73.8$; second class $1.3(36) + 27 = 73.8$ ✓
Reading it: below 36 minutes the first class's line predicts higher, above it the second class's steeper line takes over. The steeper slope means that class's scores rise faster per minute studied.

11. The mean with the outlier removed.

Drop the 90 from the list and divide by the nine scores that remain:

$$52 + 58 + 61 + 68 + 70 + 76 + 79 + 85 + 88 = 637$$

$$\text{mean} = \frac{637}{9} = 70.777\ldots$$

Rounded to the nearest hundredth, the mean is $\boxed{70.78}$ points.
What the comparison shows: removing one student out of ten moved the class mean from 72.7 down to about 70.78, nearly 2 points. One unusual value in a small data set moves a summary noticeably — which is exactly why a prediction built on a small sample deserves a stated caution.
Watch for: removing the score from the total but still dividing by ten, which gives $637 \div 10 = 63.7$ — a "mean" below every score but one, and impossible.

12. Challenge: judging the newspaper's claim.

The claim is arithmetically consistent with the line and indefensible as a statement about students. A full-credit paragraph makes at least four of these points:

- **The line does say it:** $0.8(75) + 45 = 105$, so the number is not a misreading of the line.
- **It is far outside the data:** study times run from 10 to 50 minutes. Predicting at 75 minutes is extrapolation half again beyond the largest observed value, and nothing on the plot supports the trend continuing there.
- **The prediction is impossible:** a quiz score cannot exceed 100. A model that returns 105 has been pushed past where it can be true, which is itself proof the straight line stops describing reality somewhere before 75 minutes.
- **One point already contradicts the trend:** the student who studied 22 minutes scored 90, and removing that single point moves the class mean by about 2 points. Ten students, one of them an outlier, is thin evidence for a headline.
- **"Score about" hides a causal claim:** even inside the data range, the plot shows association, not that studying longer *produces* higher scores.

What the data does support: within roughly 10 to 50 minutes of study, in this one class, longer study times went with higher quiz scores, at a rate of about 0.8 points per extra minute, with one clear exception.

Open synthesis — $\boxed{\text{open response}}$; check that the paragraph both makes a defensible positive claim and names the limits.